

Automated Chicken Counting

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Problem

Counting chickens manually in farm images is difficult and time-consuming. Chickens often overlap, move frequently, and appear at different sizes, which makes accurate counting challenging using traditional methods.

Solution

To solve this problem, a deep learning model based on a convolutional neural network (ResNet-18) is used. Instead of detecting each chicken separately, the model predicts a density map showing where chickens are likely located in the image. The total count is obtained by summing the density values.

Accuracy

The model is evaluated using metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE). Results show that the model can estimate chicken counts with relatively small errors, demonstrating that density-based approaches work well for crowded scenes.

Evaluation ---- Score: 0.8734, MAE: 4.4968, Chicken_avg: 27.5003

Final Result: {'status': True, 'score': {'public_a': 0.8729798149880961, 'private_b': 0.8734448855201352}, 'msg': 'Success!'}